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Foreword

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1 Scope

The present document specifies requirements for Rel-16 UEs supporting release independent features like:

- additional E-UTRA operating frequency bands on top of Rel-16 of TS 36.101 [2] and TS 36.133 [3];
- additional E-UTRA CA configurations (intra-band/inter-band) on top of Rel-16 of TS 36.101 [2] and TS 36.133 [3];
- additional operating bands and/or CA configurations for specific features (like UE category 0, M1, NB1);
- other release independent features (like 4Rx antenna port, high speed scenario, 8Rx antenna port).

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 36.101: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) Radio Transmission and Reception".

NOTE: The considered release is given in the text of the present document that uses [2].

[3] 3GPP TS 36.133: "Evolved Universal Terrestrial Radio Access (E-UTRA); Requirements for Support of Radio Resource Management".

[4] 3GPP TS 36.306: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) radio access capabilities".

NOTE: The considered release is given in the text of the present document that uses [4].

[5] Void

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

release independent: applicable to some frozen releases, starting from a certain release Rel-M

NOTE 1: Normally, a feature is introduced only in the latest open release Rel-N and future releases are based on the previous one so that future releases inherit the requirements of this feature. Introducing a feature "in a release independent way from Rel-M onwards" (M<N) means it was decided by TSG RAN that this feature would be also beneficial in previous, already frozen releases starting with Rel-M until Rel-(N-1). In order to avoid touching TS 36.101 [2] or TS 36.133 [3] of these frozen releases, the corresponding requirements are captured in TS 36.307 via pointers to [2] or [3] of the release in which the feature was introduced.

NOTE 2: Release independent does not mean applicable to all releases.

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

4Rx	4 UE receiver antenna ports
CA	Carrier Aggregation
CRS	Cell-specific Reference Signal
CSI	Channel State Indicator
FDD	Frequency Division Duplex
LAA	License-Assisted Access
RRC	Radio Resource Control
RRM	Radio Resource Management
SDR	Sustained Data Rate
TDD	Time Division Duplex
UE	User Equipment

3.3 Symbols

For the purposes of the present document, the following symbols apply:

N	Release in which a feature is introduced into TS 36.101 [2] or TS 36.133 [3]
M	Release from which onwards (including release M) a feature is release independent

3A Release independent features

3A.0 General

TSG-RAN has agreed for certain features (see the following clauses) to introduce them in a "release independent way".

This means for each feature:

- it is "introduced" in a release N, i.e. TS 36.101 [2] and TS 36.133 [3] of release N define certain UE requirements for this feature; the feature is indicated in the tables of the following clauses;
- it is "release independent" starting from a release M (M<N); M for the given feature is provided in the tables of the following clauses;
- UEs supporting this feature have to fulfill additional requirements in release M or higher which are specified in one or more Annexes of TS 36.307 of release N; the applicable Annexes for a given feature are provided in the tables of the following clauses.

The applicable UE Categories are specified in TS 36.306 [4] according to the release to which the UE conforms.

3A.1 Additional E-UTRA operating bands

Requirements for a Rel-16 UE for additional E-UTRA operating bands compared to TS 36.101 Rel-16 [2] are introduced via this clause.

Table 3A.1-1: E-UTRA operating bands and UE power class

Feature	Duplex-mode	Release independent from	Requirements to be fulfilled (see TS 36.307 of the release in which the band was introduced)
Operating bands, band number <= 64, Power Class 3	FDD, TDD	Rel-8	Table B.2.1-1, Table B.4.1-1
Operating bands, band number > 64, Power Class 3	FDD, TDD	Rel-9	Table B.2.1-1, Table B.4.1-1
Operating bands, NS-value > 32	FDD, TDD	Rel-10	Table B.2.1-1, Table B.4.1-1
Asymmetric operating bands, Power Class 3	FDD	Rel-10	Table B.2.1-1, Table B.4.1-1
Operating bands, band number <= 64, Power Class 1	FDD	Rel-10	Table B.2.1-1, Table B.4.1-1
Operating bands, Power Class 2	TDD	Rel-10	Table B.2.1-1, Table B.4.1-1

For example, Band 19 was introduced in the Release 9 specifications. In order to implement a UE conforming to Release 8 but supporting Band 19, it is necessary for the UE to additionally conform to some parts of the Release 9 specifications (see corresponding Annexes of TS 36.307 Rel-9 which will point to the requirements in the Rel-9 of TS 36.101 [2] or TS 36.133 [3] to be fulfilled), such as the radio frequency and radio resource management requirements for the Band 19.

3A.2 Additional E-UTRA CA configurations

Requirements for a Rel-16 UE for additional E-UTRA CA configurations compared to TS 36.101 Rel-16 [2] are introduced via this clause.

Table 3A.2-1: Intra-band contiguous CA configurations and UE CA power class

Feature	DL/UL	CA BW Class	Duplex-mode	Release independent from	requirements to be fulfilled (see 36.307 of the REL in which the CA configuration and the power class were introduced)
Intra-band contiguous CA configurations, power class 3	DL	B	FDD	Rel-10	Table B.2.2-1, Table B.3.2-1, Table B.4.2-1
		C	FDD, TDD	Rel-10	Table B.2.2-1, Table B.3.2-1, Table B.4.2-1
		D	TDD	Rel-10	Table B.2.2-1, Table B.3.2-1, Table B.4.2-1
		E	TDD	Rel-11	Table B.2.2-1, Table B.3.2-1, Table B.4.2-1
		F	TDD	Rel-12	Table B.2.2-1, Table B.3.2-1, Table B.4.2-1
	UL	B	FDD	Rel-10	Table B.2.2-1, Table B.3.2-1, Table B.4.2-1
		C, D	FDD, TDD	Rel-10	Table B.2.2-1, Table B.3.2-1, Table B.4.2-1
Intra-band contiguous CA configurations, power class 2	UL	C	TDD	Rel-10	Table B.2.2-1, Table B.3.2-1, Table B.4.2-1
NOTE1: The duplex mode "FDD, TDD" refers to a CA configuration composed by only FDD bands or only TDD bands, respectively.					

Table 3A.2-2: Inter-band CAconfigurations

Feature	DL/UL	number of bands	number of CCs	CA BW Classes	Duplex-mode	Release independent from	requirements to be fulfilled (see 36.307 of the REL in which the CA configuration was introduced)
Inter-band CA configurations	DL	2	2-4	A, B, C	FDD, TDD	Rel-10	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			2-5	D, E	FDD, TDD	Rel-11	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			2-5	A, B, C, D, E	FDD and TDD	Rel-12	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			6-7	A, C, D, E, F	FDD, TDD	Rel-14	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			6-7		FDD and TDD	Rel-14	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
		3	3	A	FDD, TDD	Rel-10	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			3-5	B, C, D	FDD, TDD	Rel-11	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			3	A	FDD and TDD	Rel-12	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			6-7	A, C, D, E, F	FDD, TDD	Rel-14	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			6-7		FDD and TDD	Rel-14	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
		4	4-5	A, C	FDD, TDD	Rel-11	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			4-5		FDD and TDD	Rel-12	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			6-7	A, C, D, E	FDD, TDD	Rel-14	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			6-7		FDD and TDD	Rel-14	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
		5	5	A	FDD, TDD	Rel-12	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			5		FDD and TDD	Rel-12	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			6-7	A, C, D	FDD, TDD	Rel-14	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
			6-7		FDD and TDD	Rel-14	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
		6	6	A	FDD	Rel-14	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
	UL	2	2-4	A, C	FDD, TDD	Rel-11	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1

			2-3	A, C	FDD and TDD	Rel-12	Table B.2.4-1, Table B.3.2-1, Table B.4.3-1 or Table B.4.4-1
NOTE1: The duplex mode "FDD, TDD" refers to a CA configuration composed by only FDD bands or only TDD bands, respectively. The duplex mode "FDD and TDD" refers to a CA configuration including both FDD and TDD bands. NOTE2: CA configurations involving downlink only operation in Band 46 are release independent from Rel-13 onwards (LAA was introduced in Rel-13). The 10 MHz channel bandwidth for Band 46 was introduced in TS 36.101 Rel-14 [2] and can be implemented in a release independent way from Rel-13.							

For example, CA configuration CA_1A-19A was introduced in the Release 11 specifications. In order to implement a UE conforming to Release 10 but supporting the CA configuration CA_1A-19A, it is necessary for the UE to additionally conform to some parts of the Release 11 specifications (see corresponding Annexes of TS 36.307 Rel-11 which will point to the requirements in the Rel-11 of TS 36.101 [2] or TS 36.133 [3] to be fulfilled), such as the radio frequency and radio resource management requirements for the CA configuration CA_1A-19A.

Table 3A.2-3: Intra-band non-contiguous CA configurations

Feature	DL/UL	number of sub-blocks	number of CCs	CA BW Classes	Duplex-mode	Release independent from	requirements to be fulfilled (see 36.307 of the REL in which the CA configuration was introduced)
Intra-band non-contiguous CA configurations	DL	2	2-5	A, C, D	FDD, TDD	Rel-11	Table B.2.3-1, Table B.3.2-1, Table B.4.5-1
		3	3-5	A, C	FDD, TDD	Rel-11	Table B.2.3-1, Table B.3.2-1, Table B.4.5-1
	UL	2	2	A	FDD	Rel-11	Table B.2.3-1, Table B.3.2-1, Table B.4.5-1
NOTE1: The duplex mode "FDD, TDD" refers to a CA configuration composed by only FDD bands or only TDD bands, respectively.							

3A.3 Additional operating bands and/or CA configurations for specific features

For a specific feature introduced in an earlier release, it may be decided in a later release to apply this specific feature in a release independent way for additional operating bands and/or CA configurations. For a Rel-16 UE corresponding requirements are then introduced via this clause.

Table 3A.3-1: Operating bands for specific features

Feature	Release independent from	Requirements to be fulfilled (see 36.307 of the REL when the feature was introduced)	Further information
Operating bands for UE category 0	Rel-12	Table B.2.9-1, Table B.3.5-1, Table B.4.10-1	Rel-14 WI LC_MTC_LTE_cat0_B25_B26-Core introduced RF, RRM, demodulation and CSI requirements for bands 25 and 26, see Table B.2.9-1, Table B.3.5-1, Table B.4.10-1
Operating bands for UE category M1	Rel-13	Table B.2.10-1, Table B.3.6-1, Table B.4.11-1	Rel-14 WI LTE_MTCe2_L1_cat1_B25_B40-Core introduced RF, RRM, demodulation and CSI requirements for bands 25 and 40, see Table B.2.10-1, Table B.3.6-1, Table B.4.11-1. Rel-15 WI LTE_bands_R15_M1_NB1-Core introduced RF, RRM, demodulation and CSI requirements for bands 14 and 71, see Table B.2.10-1, Table B.3.6-1, Table B.4.11-1. Rel-17 WI LTE_bands_R17_M1_M2_NB1_NB2 introduced RF, RRM, demodulation for band 24 and band 48, see Table B.2.10-1, Table B.3.6-1, Table B.4.11-1.
Operating bands for HD-FDD UE category M1, Power class 2	Rel-13	Table B.2.10-1, Table B.3.6-1, Table B.4.11-1	Rel-16 WI LTE_PC2_B31_B72 introduced power class 2 feature for bands 31 and 72.
Operating bands for UE category M2	Rel-14	Table B.2.11-1, Table B.3.6-1, Table B.4.11-1	Rel-15 WI LTE_bands_R15_M2_NB2-Core introduced RF and RRM requirements for bands 14 and 71, see Table B.2.11-1, Table B.4.11-1. Rel-17 WI LTE_bands_R17_M1_M2_NB1_NB2 introduced RF, RRM, demodulation for band 24 and band 48, see Table B.2.11-1, Table B.3.6-1, Table B.4.11-1.
Operating bands for HD-FDD UE category M2, Power class 2	Rel-14	Table B.2.11-1, Table B.4.11-1	Rel-16 WI LTE_PC2_B31_B72 introduced power class 2 feature for bands 31 and 72.
Operating bands for (FDD) UE category NB1	Rel-13	Table B.2.8-1, Table B.3.7-1, Table B.4.9-1	Rel-14 WI NB_IOT_R14_bands introduced RF, RRM and demodulation requirements for bands 11, 21, 25, 31, 70, see Table B.2.8-1, Table B.3.7-1, Table B.4.9-1. Rel-15 WI LTE_bands_R15_M1_NB1-Core introduced RF, RRM and demodulation for bands 4, 14 and 71 see Table B.2.8-1, Table B.3.7-1, Table B.4.9-1. Rel-16 WI LTE_bands_R16_M1_NB1 introduced RF, RRM, demodulation for band 65, see Table B.2.8-1, Table B.3.7-1, Table B.4.9-1. Rel-17 WI LTE_bands_R17_M1_M2_NB1_NB2 introduced RF, RRM, demodulation for band 24, see Table B.2.8-1, Table B.3.7-1, Table B.4.9-1.
Operating bands (FDD) for UE category NB2	Rel-14	Table B.2.12-1, Table 3.7-1, Table B.4.9-1	Rel-15 WI LTE_bands_R15_M2_NB2-Core introduced RF, RRM and demodulation requirements for bands 4, 14 and 71, see Table B.2.12-1, Table 3.7-1, Table B.4.9-1. Rel-16 WI LTE_bands_R16_M2_NB2 introduced RF, RRM, demodulation for band 65 , see Table B.2.12-1, Table B.3.7-1, Table B.4.9-1. Rel-17 WI LTE_bands_R17_M1_M2_NB1_NB2 introduced RF, RRM, demodulation for band 24, see Table B.2.8-1, Table B.3.7-1, Table B.4.9-1.
Operating bands (TDD) for UE category NB1 and NB2	Rel-15	Table B.2.12-1, Table 3.7-1, Table B.4.9-1	Rel-16 WI LTE_bands_R16_M1_NB1 and Rel-16 WI LTE_bands_R16_M2_NB2 introduced RF, RRM, demodulation for band 42 and band 43, see Table B.2.12-1, Table B.3.7-1, Table B.4.9-1. Rel-17 WI LTE_bands_R17_M1_M2_NB1_NB2 introduced RF, RRM, demodulation for band 48, see Table B.2.12-1, Table B.3.7-1, Table B.4.9-1.
Operating bands for UE category 1bis	Rel-13	Table B.2.15-1, Table B.3.8-1, Table B.4.14-1	Rel-16 RF, RRM, demodulation and CSI requirements for band 34 and band 40, see Table B.2.15-1, Table B.3.8-1, Table B.4.14-1.

Table 3A.3-2: CA configurations for specific features

Feature	Release independent from	Requirements to be fulfilled (see 36.307 of the REL when the feature was introduced)	Further information
Operating bands for V2X communication with con-current operation	Rel-14	Table B.2.13-1, Table B.4.12-1	Rel-15 WI V2X new band combinations (V2X_5A-47A, V2X_20A-47A, V2X_34A-47A, V2X_28A-47A, V2X_71A-47A) introduced and should be satisfied for the RF and RRM requirements in Table B.2.13-1, Table B.4.12-1
Operating band for V2X communication with multi-carrier at Band 47	Rel-14	Table B.2.13-1, Table B.4.12-1	In Rel-15 WI for eV2X, introduce intra-band multi-carrier V2X_47C and V2X_47C1 and should be satisfied for the RF and RRM requirements in Table B.2.13-1, Table B.4.12-1

3A.4 Other release independent features

This clause covers requirements for a Rel-16 UE coming from all other release independent features that are not covered under clause 3A.1, 3A.2 and 3A.3, e.g. generic baseband requirements or requirements that are not band/CA configuration specific.

Table 3A.4-1: Additional requirements of other release independent features

Feature	Release independent from	Requirements to be fulfilled (see 36.307 of the REL when the feature was introduced)	Further information
RF and performance requirements for 4Rx UEs	Rel-10	Table C.1-1, Table C.2-1 for single carrier and Table C.1-2, Table C.2-2 for CA	REL-13 WI LTE_4Rx_AP_DL introduced: - single carrier RF requirements for bands 1, 2, 3, 7, 20, 39, 41, 42: see Table C.1-1 - CA RF requirements for CA_3A-42A and other 1UL CA configurations (see TS 36.101 REL-13 [2] Table 7.3.1A-0a NOTE 20): see Table C.1-2 - single carrier performance requirements for demodulation and CSI: see Table C.2-1 REL-14 WI LTE_4Rx_AP_DL_bands introduced: - single carrier RF requirements for band 35, 40: see Table C.1-1 - CA RF requirements for some further 1UL CA configurations (see TS 36.101 REL-14 [2]): see Table C.1-2 REL-14 WI LTE_4Rx_AP_DL_CA introduced: - CA RF requirements for some 2DL/2UL CA configurations (see TS 36.101 REL-14 [2]): see Table C.1-2 - CA performance requirements for demodulation/SDR and CSI: see Table C2-2 REL-15 WI LTE_4Rx_AP_DL_bands_R15 introduced: - single carrier RF requirements for band 4, 34, 43, 66: see Table C.1-1 - CA RF requirements for some further 1UL CA configurations (see TS 36.101 REL-15 [2]): see Table C.1-2
RF and performance requirements for 8Rx UEs	Rel-13	Table E.1-1, Table E.2-1 for single carrier and Table E.1-2, Table E.2-2 for CA	REL-15 WI LTE_8Rx_AP_DL introduced: - single carrier RF requirements for band 41, 42,43: see Table E.1-1 - CA RF requirements for CA_41C, CA_42C and CA_41A-42A CA configurations (see TS 36.101 REL-15 [2]): see Table E.1-2 - single carrier performance requirements for demodulation and CSI: see Table E.2-1 - CA performance requirements for demodulation/SDR: see Table E.2-2
RRM and demodulation requirements for high speed scenario	Rel-13 (NOTE 1)	Table D.1-1, Table D.2-1	Rel-14 WI LTE_high_speed introduced band independent RRM and demodulation requirements. see Table D.1-1, Table D.2-1
NOTE 1: Rel-13 UEs supporting the high speed scenario are assumed to read the Rel-14 high speed scenario information, which is broadcast to all UEs.			

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Annex A (informative) : Frequency arrangement for overlapping operating bands

The following information is provided in order to assist a UE derive the DL EARFCN and UL EARFCN in a multi-band environment, in which multiple overlapping operating bands may be indicated in the fields *freqBandIndicator* and *multiBandInfoList* of SIB1.

The overlapping bands, independent of release, which may be indicated in a cell are shown in Table A-1 for applicable E-UTRA bands. The DL EARFCN and UL EARFCN are derived according to TS 36.101 Rel-16 [2].

Table A-1: Overlapping bands (multi-band environments) for each E-UTRA band

E-UTRA Operating Band	Overlapping E-UTRA operating bands	Duplex Mode
2	25	FDD
3	9	FDD
4	10, 66	FDD
5	18, 19, 26	FDD
9	3	FDD
10	4, 66	FDD
12	17	FDD
17	12	FDD
18	5, 26, 27	FDD
19	5, 26	FDD
25	2	FDD
26	5, 18, 19, 27	FDD
27	18, 26	FDD
33	39	TDD
38	41	TDD
39	33	TDD
41	38	TDD
66	4, 10	FDD

Annex B (normative): Common Requirements for bands or CA

B.1 Purpose of annex

The purpose of Annex B is to group the requirements that are common for several bands or CA configurations in this specification and use the common tables as references.

B.2 Common RRM requirements

B.2.1 Common RRM requirements for a release independent band

The requirements and test cases listed in Table B.2.1-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.1-1: Common RRM requirements for a release independent band

Clause	Description
4 Note 1	E-UTRAN RRC_IDLE state mobility
5	E-UTRAN RRC_CONNECTED state mobility
6 Note 2	RRC Connection Mobility Control
7 Note 3	Timing and signalling characteristics
8 Note 4	UE Measurements Procedures in RRC_CONNECTED State
9 Note 5	Measurements performance requirements for UE
A.4 Note 1	E-UTRAN RRC_IDLE state
A.5	E-UTRAN RRC CONNECTED Mode Mobility
A.6 Note 2	RRC Connection Control
A.7 Note 3	Timing and Signalling Characteristics
A.8 Note 4	UE Measurements Procedures
A.9 Note 5	Measurement Performance Requirements

NOTE 1: All requirements and the corresponding test cases shall apply, except:
- for supporting the corresponding band in Rel-9 and below: clause 4.3 (Minimization of Drive Tests).

NOTE 2: All requirements and the corresponding test cases shall apply, except:
- for supporting the corresponding band in Rel-8: clauses 6.3 (RRC Connection Release with Redirection), 6.4 (CSG Proximity Indication for E-UTRAN and UTRAN).

NOTE 3: All requirements and corresponding test cases shall apply, except those defined in clauses 7.4 and 7.5.

NOTE 4: All requirements and corresponding test cases shall apply, except:
- for supporting the corresponding band in Rel-8: clauses 8.1.2.5 (E-UTRAN OTDOA Intra-Frequency RSTD Measurements), 8.1.2.6 (E-UTRAN Inter-Frequency OTDOA Measurements), 8.1.2.7 (E-UTRAN E-CID Measurements).

NOTE 5: All requirements and corresponding test cases shall apply, except:
- for supporting the corresponding band in Rel-8: clauses 9.1.9 (UE Rx–Tx time difference), 9.1.10 (Reference Signal Time Difference).
- for supporting the corresponding band in Rel-11 or below: the RSRP absolute accuracy requirement under normal conditions in table 9.1.2.1-1, 9.1.2.3-1, 9.1.2.5-1 and 9.1.3.1-1 when Io≤-70dBm is ±6dB.
- for supporting the corresponding band in Rel-11 or below: the interfrequency RSRP relative accuracy requirement under normal conditions in table 9.1.3.2-1 is ±6dB.

NOTE 6: In addition to the exceptions above, all requirements and test cases in this table shall apply, except those defined for:
- carrier aggregation;
- for supporting the corresponding band in Rel-9 or below: measurements under time-domain measurement resource restriction without CRS assistance information;
- for supporting the corresponding band in Rel-10 or below: measurements under time-domain measurement resource restriction with CRS assistance information;
- for supporting the corresponding band in Rel-11 or below: requirements introduced in Rel-12.

B.2.2 Common RRM requirements for an intra-band contiguous CA configuration

The requirements and test cases listed in Table B.2.2-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.2-1: Common RRM requirements for a release independent single-band CA configuration

Clause	Description
7.1	UE transmit timing
7.7	SCell Activation and Deactivation Delay for E-UTRA Carrier Aggregation
7.8	Interruptions with Carrier Aggregation
8.2	Capabilities for Support of Event Triggering and Reporting Criteria
8.3	Measurements for E-UTRA carrier aggregation
8.4	OTDOA RSTD Measurements for E-UTRAN carrier aggregation
9.1.11 Note 3	Carrier aggregation measurement accuracy
9.1.12	Reference Signal Time Difference (RSTD) Measurement Accuracy Requirements for Carrier Aggregation
A.7	Timing and Signalling Characteristics
A.8	UE Measurements Procedures
A.9 Note 3	Measurement Performance Requirements
NOTE 1: Only requirements and test cases defined for intra-band contiguous carrier aggregation shall apply. NOTE 2: In addition to the exceptions above, all requirements and test cases in this table shall apply, except: - for supporting the corresponding band in Rel-11 or below: requirements introduced in Rel-12. NOTE 3: - For supporting the corresponding band in Rel-11 or below: the RSRP absolute accuracy requirement under normal conditions in table 9.1.2.1-1, 9.1.2.3-1, 9.1.2.5-1 and 9.1.3.1-1 when Io≤-70dBm is ±6dB. - For supporting the corresponding band in Rel-11 or below: the interfrequency RSRP relative accuracy requirement under normal conditions in table 9.1.3.2-1 is ±6dB.	

B.2.3 Common RRM requirements for an intra-band non-contiguous CA with single uplink configuration

The requirements and test cases listed in Table B.2.3-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.3-1: Common RRM requirements for a release independent single-band CA configuration

Clause	Description
7.1	UE transmit timing
7.7	SCell Activation and Deactivation Delay for E-UTRA Carrier Aggregation
7.8	Interruptions with Carrier Aggregation
8.2	Capabilities for Support of Event Triggering and Reporting Criteria
8.3	Measurements for E-UTRA carrier aggregation
8.4	OTDOA RSTD Measurements for E-UTRAN carrier aggregation
9.1.11 Note 3	Carrier aggregation measurement accuracy
9.1.12	Reference Signal Time Difference (RSTD) Measurement Accuracy Requirements for Carrier Aggregation
A.7	Timing and Signalling Characteristics
A.8	UE Measurements Procedures
A.9 Note 3	Measurement Performance Requirements
NOTE 1: Only requirements and test cases defined for intra-band non-contiguous carrier aggregation with single uplink shall apply. NOTE 2: In addition to the exceptions above, all requirements and test cases in this table shall apply, except: - for supporting the corresponding band in Rel-11 or below: requirements introduced in Rel-12. NOTE 3: - For supporting the corresponding band in Rel-11 or below: the RSRP absolute accuracy requirement under normal conditions in table 9.1.2.1-1, 9.1.2.3-1, 9.1.2.5-1 and 9.1.3.1-1 when Io≤-70dBm is ±6dB - for supporting the corresponding band in Rel-11 or below: the interfrequency RSRP relative accuracy requirement under normal conditions in table 9.1.3.2-1 is ±6dB.	

B.2.4 Common RRM requirements for an inter-band CA with single uplink configuration

The requirements and test cases listed in Table B.2.4-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.4-1: Common RRM requirements for a release independent band-combination CA configuration

Clause	Description
7.1	UE transmit timing
7.7	SCell Activation and Deactivation Delay for E-UTRA Carrier Aggregation
7.8	Interruptions with Carrier Aggregation
8.2	Capabilities for Support of Event Triggering and Reporting Criteria
8.3	Measurements for E-UTRA carrier aggregation
8.4	OTDOA RSTD Measurements for E-UTRAN carrier aggregation
9.1.11 Note 3	Carrier aggregation measurement accuracy
9.1.12	Reference Signal Time Difference (RSTD) Measurement Accuracy Requirements for Carrier Aggregation
A.7	Timing and Signalling Characteristics
A.8	UE Measurements Procedures
A.9 Note 3	Measurement Performance Requirements
NOTE 1: Only requirements and test cases defined for inter-band with single uplink carrier aggregation shall apply. NOTE 2: In addition to the exceptions above, all requirements and test cases in this table shall apply, except: - for supporting the corresponding band in Rel-11 or below: requirements introduced in Rel-12. NOTE 3: - For supporting the corresponding band in Rel-11 or below: the RSRP absolute accuracy requirement under normal conditions in table 9.1.2.1-1, 9.1.2.3-1, 9.1.2.5-1 and 9.1.3.1-1 when Io≤-70dBm is ±6dB. - for supporting the corresponding band in Rel-11 or below: the interfrequency RSRP relative accuracy requirement under normal conditions in table 9.1.3.2-1 is ±6dB.	

B.2.5 Common RRM requirements for an inter-band CA with dual uplink configuration

The requirements and test cases listed in Table B.2.5-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.5-1: Common RRM requirements for a release independent band-combination CA configuration with dual uplink

Clause	Description
7.1	UE transmit timing
7.7	SCell Activation and Deactivation Delay for E-UTRA Carrier Aggregation
7.8	Interruptions with Carrier Aggregation
7.17	Maximum Transmission Timing Difference in Dual Connectivity
8.2	Capabilities for Support of Event Triggering and Reporting Criteria
8.3	Measurements for E-UTRA carrier aggregation
8.4	OTDOA RSTD Measurements for E-UTRAN carrier aggregation
9.1.11 Note 3	Carrier aggregation measurement accuracy
9.1.12	Reference Signal Time Difference (RSTD) Measurement Accuracy Requirements for Carrier Aggregation
A.7	Timing and Signalling Characteristics
A.8	UE Measurements Procedures
A.9 Note 3	Measurement Performance Requirements
NOTE 1: Only requirements and test cases defined for inter-band with dual uplink carrier aggregation shall apply. NOTE 2: In addition to the exceptions above, all requirements and test cases in this table shall apply, except: - for supporting the corresponding band in Rel-11 or below: requirements introduced in Rel-12. NOTE 3: - For supporting the corresponding band in Rel-11 or below: the RSRP absolute accuracy requirement under normal conditions in table 9.1.2.1-1, 9.1.2.3-1, 9.1.2.5-1 and 9.1.3.1-1 when Io≤-70dBm is ±6dB. - for supporting the corresponding band in Rel-11 or below: the interfrequency RSRP relative accuracy requirement under normal conditions in table 9.1.3.2-1 is ±6dB.	

B.2.6 Common RRM requirements for an intra-band non-contiguous CA with dual uplink configuration

The requirements and test cases listed in Table B.2.6-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.6-1: Common RRM requirements for a release independent single-band CA configuration with dual uplink

Clause	Description
7.1	UE transmit timing
7.7	SCell Activation and Deactivation Delay for E-UTRA Carrier Aggregation
7.8	Interruptions with Carrier Aggregation
7.17	Maximum Transmission Timing Difference in Dual Connectivity
8.2	Capabilities for Support of Event Triggering and Reporting Criteria
8.3	Measurements for E-UTRA carrier aggregation
8.4	OTDOA RSTD Measurements for E-UTRAN carrier aggregation
9.1.11 Note 3	Carrier aggregation measurement accuracy
9.1.12	Reference Signal Time Difference (RSTD) Measurement Accuracy Requirements for Carrier Aggregation
A.7	Timing and Signalling Characteristics
A.8	UE Measurements Procedures
A.9 Note 3	Measurement Performance Requirements
NOTE 1: Only requirements and test cases defined for intra-band non-contiguous carrier aggregation with dual uplinks shall apply. NOTE 2: In addition to the exceptions above, all requirements and test cases in this table shall apply, except: - for supporting the corresponding band in Rel-11 or below: requirements introduced in Rel-12. NOTE 3: - For supporting the corresponding band in Rel-11 or below: the RSRP absolute accuracy requirement under normal conditions in table 9.1.2.1-1, 9.1.2.3-1, 9.1.2.5-1 and 9.1.3.1-1 when Io≤-70dBm is ±6dB. - for supporting the corresponding band in Rel-11 or below: the interfrequency RSRP relative accuracy requirement under normal conditions in table 9.1.3.2-1 is ±6dB.	

B.2.7 Common RRM requirements for an inter-band CA with three uplink configuration

The requirements and test cases listed in Table B.2.7-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.7-1: Common RRM requirements for a release independent band-combination CA configuration with three uplink

Clause	Description
7.1	UE transmit timing
7.7	SCell Activation and Deactivation Delay for E-UTRA Carrier Aggregation
7.8	Interruptions with Carrier Aggregation
7.17	Maximum Transmission Timing Difference in Dual Connectivity
8.2	Capabilities for Support of Event Triggering and Reporting Criteria
8.3	Measurements for E-UTRA carrier aggregation
8.4	OTDOA RSTD Measurements for E-UTRAN carrier aggregation
9.1.11 Note 3	Carrier aggregation measurement accuracy
9.1.12	Reference Signal Time Difference (RSTD) Measurement Accuracy Requirements for Carrier Aggregation
NOTE 1: Only requirements defined for three uplink carrier aggregation shall apply. There are no test cases defined with a three uplink carrier aggregation configuration. NOTE 2: In addition to the exceptions above, all requirements and test cases in this table shall apply, except: - for supporting the corresponding band in Rel-11 or below: requirements introduced in Rel-12. NOTE 3: - For supporting the corresponding band in Rel-11 or below: the RSRP absolute accuracy requirement under normal conditions in table 9.1.2.1-1, 9.1.2.3-1, 9.1.2.5-1 and 9.1.3.1-1 when Io≤-70dBm is ±6dB. - for supporting the corresponding band in Rel-11 or below: the interfrequency RSRP relative accuracy requirement under normal conditions in table 9.1.3.2-1 is ±6dB.	

B.2.8 Common RRM requirements for operating bands for UE category NB1

The requirements and test cases listed in Table B.2.8-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.8-1: Common RRM requirements for release independent operating bands for UE category NB1

Clause	Description
4.6	Cell Selection and Reselection Requirements for UE category NB1
6.6	Random Access for UE category NB1
7.23	Radio Link Monitoring for category NB1 UE
8.14	Measurements for UE category NB1
9.1.22	Measurement accuracy for UE Category NB1
9.1.23	Power Headroom for UE category NB1
NOTE 1: Only requirements and test cases defined for UE category NB1 shall apply.	

B.2.9 Common RRM requirements for operating bands for UE category 0

The requirements and test cases listed in Table B.2.9-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.9-1: Common RRM requirements for release independent operating bands for a UE category 0

Clause	Description
7.11	Radio Link Monitoring for UE category 0
8.5	Measurements for UE category 0
9.1.13	Measurement accuracy for UE category 0

B.2.10 Common RRM requirements for operating bands for UE category M1

The requirements and test cases listed in Table B.2.10-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.10-1: Common RRM requirements for release independent operating bands for a UE category M1

Clause	Description
4.7	Cell Selection and Re-selection Requirements for UE category M1
5.5	E-UTRAN Handover for cat.M1 UEs in CEModeA
5.6	E-UTRAN Handover for cat.M1 UEs in CEModeB
6.2.3	Random Access Requirements for cat.M1 UEs
6.7	RRC Re-establishment for cat.M1 UEs
6.8	RRC Connection Release with Redirection for Cat-M1 UEs
7.19	Radio Link Monitoring for UE Category M1
7.24	UE transmit timing for category M1
7.27	UE timer accuracy for category M1
7.28	Timing Advance for Category M1
8.13	Measurements for UE category M1
9.1.21	Measurement accuracy for UE category M1

B.2.11 Common RRM requirements for operating bands for UE category M2

The requirements and test cases listed in Table B.2.11-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.11-1: Common RRM requirements for release independent operating bands for a UE category M2

Clause	Description
4.7	Cell Selection and Re-selection Requirements
5.5	E-UTRAN Handover in CEModeA
5.6	E-UTRAN Handover in CEModeB
6.2.3	Random Access Requirements
6.7	RRC Re-establishment
6.8	RRC Connection Release with Redirection
7.19	Radio Link Monitoring
7.26	UE transmit timing for category M2
7.27	UE timer accuracy
7.28	Timing Advance
8.13.2.1 and 8.13.3.1	E-UTRAN intra frequency measurement requirements
8.13.2.6 and 8.13.3.5	E-UTRAN inter frequency measurement requirements
8.13.2.7 and 8.13.3.6	UE measurement capability
8.13.2.5.1, 8.13.2.5.2, 8.13.2.5.3, 8.13.2.5.4, 8.13.2.5.5, 8.13.2.5.6 and 8.13.3.4	E-UTRAN E-CID measurements requirements
8.16	Measurements for UE Category M2
9.1.21	Measurement accuracy
9.1.25	Measurement accuracy for UE category M2

B.2.12 Common RRM requirements for operating bands for UE category NB2

The requirements and test cases listed in Table B.2.12-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.12-1: Common RRM requirements for release independent operating bands for UE category NB2

Clause	Description
4.6.1 and 4.6.2	Cell selection and re-selection requirements
4.8	UE Positioning measurement in idle state
6.5	RRC Re-establishment requirements
6.6	Random access requirements
6.9	RRC connection redirection to non-anchor carrier requirements
7.20	UE transmit timing requirements
7.21	UE timer accuracy requirements
7.22	Timing advance requirements
7.23	Radio link monitoring requirements
8.14	UE RRC_CONNECTED state measurement requirement
9.1.22	UE measurement accuracy requirements
9.1.23	Power headroom requirements
NOTE 1: Only requirements and test cases defined for UE category NB2 shall apply.	

B.2.13 Common RRM requirements for operating bands for LTE-based V2X Communication

The requirements and test cases listed in Table B.2.13-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.13-1: Common RRM requirements for release independent operating bands for LTE-based V2X communication

Clause	Description
13.2	UE Transmit Timing
13.3	Initiation/Cease of SLSS Transmissions
13.4	Selection / Reselection of V2X Synchronization Reference Source
13.5	Autonomous Resource Selection/Reselection measurements
13.6	Congestion Control measurements
13.7	Interruption
13.8	Reliability of GNSS signal

B.2.14 Common RRM requirements for an inter-band CA with four uplink configuration

The requirements and test cases listed in Table B.2.14-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.14-1: Common RRM requirements for a release independent band-combination CA configuration with four uplink

Clause	Description
7.1	UE transmit timing
7.7	SCell Activation and Deactivation Delay for E-UTRA Carrier Aggregation
7.8	Interruptions with Carrier Aggregation
7.17	Maximum Transmission Timing Difference in Dual Connectivity
8.2	Capabilities for Support of Event Triggering and Reporting Criteria
8.3	Measurements for E-UTRA carrier aggregation
8.4	OTDOA RSTD Measurements for E-UTRAN carrier aggregation
9.1.11 Note 3	Carrier aggregation measurement accuracy
9.1.12	Reference Signal Time Difference (RSTD) Measurement Accuracy Requirements for Carrier Aggregation
NOTE 1: Only requirements defined for four uplink carrier aggregation shall apply. There are no test cases defined with a four uplink carrier aggregation configuration. NOTE 2: In addition to the exceptions above, all requirements and test cases in this table shall apply, except: - for supporting the corresponding band in Rel-11 or below: requirements introduced in Rel-12. NOTE 3: For supporting the corresponding band in Rel-11 or below: the RSRP absolute accuracy requirement under normal conditions in table 9.1.2.1-1, 9.1.2.3-1, 9.1.2.5-1 and 9.1.3.1-1 when Io≤-70dBm is ±6dB. For supporting the corresponding band in Rel-11 or below: the interfrequency RSRP relative accuracy requirement under normal conditions in table 9.1.3.2-1 is ±6dB.	

B.2.15 Common RRM requirements for operating bands for UE category 1bis

The requirements and test cases listed in Table B.2.15-1 are specified in TS 36.133 Rel-16 [3].

Table B.2.15-1: Common RRM requirements for release independent operating bands for a UE category 1bis

Clause	Description
8.1.2.5.3, 8.1.2.5.4	Intra-Frequency OTDOA Measurements for UE Category 1bis
8.1.2.6.5, 8.1.2.6.6, 8.1.2.6.7, 8.1.2.6.8	Inter-Frequency OTDOA Measurements for UE Category 1bis
9.1.2.7, 9.1.2.8	Intra-frequency RSRP Accuracy Requirements for UE Category 1bis
9.1.3.1, 9.1.3.4	Intra-frequency RSRP Accuracy Requirements for UE Category 1bis
9.1.5.5	Intra-frequency RSRQ Accuracy Requirements for UE Category 1bis
9.1.6.5, 9.1.6.6	Inter-frequency RSRQ Accuracy Requirements for UE Category 1bis
9.1.10.5, 9.1.10.6	Reference Signal Time Difference (RSTD) for UE Category 1bis

B.3 Common UE performance requirements

B.3.1 Void

B.3.2 Common UE performance requirements and tests for different CA configurations and combination sets

The requirements and test cases listed in Table B.3.2-1 are specified in TS 36.101 Rel-16 [2].

Table B.3.2-1: Common UE performance requirements and tests for different CA configurations and combination sets

Clause	Description
8.2.1.1.1	Single-antenna port performance (FDD)
8.2.2.1.1	Single-antenna port performance (TDD)
8.2.3.1.1	Single-antenna port performance (TDD-FDD CA)
8.2.1.3.1	Open-loop spatial multiplexing performance - Minimum Requirement 2 Tx Antenna Port (FDD)
8.2.2.3.1	Open-loop spatial multiplexing performance - Minimum Requirement 2 Tx Antenna Port (TDD)
8.2.3.3.1	Open-loop spatial multiplexing performance - Minimum Requirement 2 Tx Antenna Port (TDD-FDD CA)
8.2.1.3.1A	Open-loop spatial multiplexing performance - Soft buffer management test (FDD)
8.2.2.3.1A	Open-loop spatial multiplexing performance - Soft buffer management test (TDD)
8.2.3.3.1A	Open-loop spatial multiplexing performance - Soft buffer management test (TDD-FDD CA)
8.2.1.4.3	Closed-loop spatial multiplexing performance - Minimum Requirement Multi-Layer Spatial Multiplexing 4 Tx Antenna Port (FDD)
8.2.2.4.3	Closed-loop spatial multiplexing performance - Minimum Requirement Multi-Layer Spatial Multiplexing 4 Tx Antenna Port (TDD)
8.2.3.4.3	Closed-loop spatial multiplexing performance - Minimum Requirement Multi-Layer Spatial Multiplexing 4 Tx Antenna Port (TDD-FDD CA)
8.2.1.7	Carrier aggregation with power imbalance (FDD)
8.2.1.8	Intra-band non-contiguous carrier aggregation with timing offset (FDD)
8.2.2.7	Carrier aggregation with power imbalance (TDD)
8.7.1	Sustained downlink data rate provided by lower layers (FDD)
8.7.2	Sustained downlink data rate provided by lower layers (TDD)
8.7.5	Sustained downlink data rate provided by lower layers (TDD-FDD CA)
8.7.12.1	Sustained downlink data rate provided by lower layers (FDD CA in licensed bands)
8.7.12.2	Sustained downlink data rate provided by lower layers (TDD CA in licensed bands)
8.7.12.3	Sustained downlink data rate provided by lower layers (TDD-FDD CA in licensed bands)
9.6.1.1	Additional requirements for carrier aggregation - Periodic reporting on multiple cells (Cell Specific Reference symbols) (FDD)
9.6.1.2	Additional requirements for carrier aggregation - Periodic reporting on multiple cells (Cell Specific Reference symbols) (TDD)
9.6.1.3	Additional requirements for carrier aggregation - Periodic reporting on multiple cells (Cell Specific Reference symbols) (TDD-FDD CA)
NOTE 1: The applicability of requirements for different CA configurations and bandwidth combination sets is specified in Clause 8.1.2.3 and 9.1.1.2. NOTE 2: The test coverage for different number of component carriers is defined in 8.1.2.4.	

B.3.3 Void

B.3.4 Void

B.3.5 Common UE performance requirements and tests for operating bands for UE category 0

The requirements and test cases listed in Table B.3.5-1 are specified in TS 36.101 Rel-16 [2].

Table B.3.5-1: Common UE performance requirements and tests for release independent operating bands for UE category 0

Clause	Description
8.9	Demodulation (single receiver antenna)
9.7	CSI reporting (Single receiver antenna)

B.3.6 Common UE performance requirements and tests for operating bands for UE category M1 and M2

The requirements and test cases listed in Table B.3.6-1 are specified in TS 36.101 Rel-16 [2].

Table B.3.6-1: Common UE performance requirements and tests for release independent operating bands for UE category M1 and M2

Clause	Description
8.11	Demodulation (UE supporting coverage enhancement)
9.8	CSI reporting (UE supporting coverage enhancement)

B.3.7 Common UE performance requirements and tests for operating bands for UE category NB1 and NB2

The requirements and test cases listed in Table B.3.7-1 are specified in TS 36.101 Rel-16 [2].

Table B.3.7-1: Common UE performance requirements and tests for release independent operating bands for UE category NB1 and NB2

Clause	Description
8.12	Demodulation of Narrowband IoT

B.3.8 Common UE performance requirements and tests for operating bands for UE category 1bis

The requirements and test cases listed in Table B.3.8-1 are specified in TS 36.101 Rel-16 [2].

Table B.3.8-1: Common UE performance requirements and tests for release independent operating bands for UE category 1bis

Clause	Description
9.7.1.3, 9.7.1.4, 9.7.2.3, 9.7.2.4	CSI reporting (Single receiver antenna)

B.4 Common UE RF requirements

B.4.1 Common UE RF requirements for a release independent band

The requirements and test cases listed in Table B.4.1-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.1-1: Common UE RF requirements for a release independent band

Clause	Description
5.5	Operating bands
5.6	Channel bandwidth
5.7	Channel arrangement
6.2	Transmit power
6.3	Output power dynamics
6.5	Transmit signal quality
6.6	Output RF spectrum emissions
6.7	Transmit intermodulation
7.3	Reference sensitivity power level
7.4	Maximum input level
7.5	Adjacent Channel Selectivity (ACS)
7.6	Blocking characteristics
7.7	Spurious response
7.8	Intermodulation characteristics
7.9	RX spurious emissions

B.4.2 Common UE RF requirements for an intra-band contiguous CA configuration

The requirements and test cases listed in Table B.4.2-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.2-1: Common UE RF requirements for a release independent intra-band contiguous CA configuration

Clause	Description
5.5A	Operating bands for CA
5.6A	Channel bandwidths per operating band for CA
5.7.1A	Channel spacing for CA
5.7.2A	Channel raster for CA
5.7.4A	TX–RX frequency separation for CA
6.2.2A	UE maximum output power for CA
6.2.3A	UE maximum output power for modulation/channel bandwidth for CA
6.2.4A	UE maximum output power with additional requirements for CA
6.2.5A	Configured transmitted power for CA
6.3.2A	UE Minimum utput power for CA
6.3.3A	UE Trasnsmit OFF power for CA
6.3.4A	ON/OFF time mask for CA
6.3.5A	Power control for CA
6.5.1A	Frequency error for CA
6.5.2A	Transmit modulation quality for CA
6.6.1A	Occupied bandwidth for CA
6.6.2.1A	Spectrum emission mask for CA
6.6.2.2A	Additional Spectrum Emission mask for CA
6.6.2.3.2A	UTRA ACLR for CA
6.6.2.3.3A	E-UTRA ACLR for CA
6.6.3.1A	Minimum requirements for CA
6.6.3.2A	Spurious emission band UE co-existence for CA
6.6.3.3A	Additional spurious emissions for CA
6.7.1A	Minimum requirement for CA
7.3.1A	Reference sensitivity for CA
7.4.1A	Maximum input level for CA
7.5.1A	Adjacent Channel Selectivity (ACS) for CA
7.6.1.1A	In-band blocking for CA
7.6.2.1A	Out-of-band blocking for CA
7.6.3.1A	Narrow band blocking for CA
7.7.1A	Spurious response for CA
7.8.1A	Wideband intermodulation for CA
7.10.1A	Receiver response for CA

B.4.3 Common UE RF requirements for an single uplink inter-band CA configuration

The requirements and test cases listed in Table B.4.3-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.3-1: Common UE RF requirements for a release independent inter-band CA configuration

Clause	Description
5.5A	Operating bands for CA
5.6A.1	Channel bandwidths per operating band for CA
5.7.2A	Channel raster for CA
6.2.2A	UE maximum output power for CA
6.2.3A	UE maximum output power for modulation/channel bandwidth for CA
6.2.5	Configured transmitted power
7.3.1A	Reference sensitivity for CA
7.4.1A	Maximum input level for CA
7.5.1A	Adjacent Channel Selectivity (ACS) for CA
7.6.1.1A	In-band blocking for CA
7.6.2.1A	Out-of-band blocking for CA
7.6.3.1A	Narrow band blocking for CA
7.7.1A	Spurious response for CA
7.8.1A	Wideband intermodulation for CA

B.4.4 Common UE RF requirements for an inter-band CA configuration including an operating band without uplink band

The requirements and test cases listed in Table B.4.4-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.4-1: Common UE RF requirements for a release independent inter-band CA configuration including an operating band without uplink band

Clause	Description
5.5	Operating bands
5.5A	Operating bands for CA
5.6A.1	Channel bandwidths per operating band for CA
5.7	Channel arrangement
6.2.2A	UE maximum output power for CA
6.2.3A	UE maximum output power for modulation/channel bandwidth for CA
6.2.5	Configured transmitted power
7.3.1A	Reference sensitivity for CA
7.4.1A	Maximum input level for CA
7.5.1A	Adjacent Channel Selectivity (ACS) for CA
7.6.1.1A	In-band blocking for CA
7.6.2.1A	Out-of-band blocking for CA
7.6.3.1A	Narrow band blocking for CA
7.7.1A	Spurious response for CA
7.8.1A	Wideband intermodulation for CA

B.4.5 Common UE RF requirements for a single uplink intra-band non-contiguous CA configuration

The requirements and test cases listed in Table B.4.5-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.5-1: Common UE RF requirements for a release independent single uplink intra-band non-contiguous CA configuration

Clause	Description
5.5A	Operating bands for CA
5.6A1	Channel bandwidths per operating band for CA
5.7.2A	Channel raster for CA
6.2.2A	UE maximum output power for CA
6.2.3A	UE maximum output power for modulation/channel bandwidth for CA
7.3.1A	Reference sensitivity for CA
7.4.1A	Maximum input level for CA
7.5.1A	Adjacent Channel Selectivity (ACS) for CA
7.6.1.1A	In-band blocking for CA
7.6.2.1A	Out-of-band blocking for CA
7.6.3.1A	Narrow band blocking for CA
7.7.1A	Spurious response for CA
7.8.1A	Wideband intermodulation for CA

B.4.6 Common UE RF requirements for dual uplink inter-band CA configuration

The requirements and test cases listed in Table B.4.6-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.6-1: Common UE RF requirements for a release independent dual uplink inter-band CA configuration

Clause	Description
5.6A.1	Channel bandwidths per operating band for CA
6.2.2A	UE maximum output power for CA
6.2.5A	Configured transmitted Power for CA
6.3.2A	UE Minimum output power for CA
6.3.3A	UE Transmit OFF power for CA
6.3.4A	ON/OFF time mask for CA
6.3.5A	Power control for CA
6.5.1A	Frequency error for CA
6.5.2A	Transmit modulation quality for CA
6.6.1A	Occupied bandwidth for CA
6.6.2.1A	Spectrum emission mask for CA
6.6.2.3	Adjacent Channel Leakage Ratio
6.6.3.1A	Spurious Emission for CA
6.6.3.2A	Spurious emission band UE co-existence for CA
6.7.1A	Transmit intermodulation for CA
7.3.1A	Reference sensitivity for CA
7.6.2.1A	Out-of-band blocking for CA
7.7.1A	Spurious response for CA

B.4.7 Common UE RF requirements for dual uplink intra-band non-contiguous CA configuration

The requirements and test cases listed in Table B.4.7-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.7-1: Common UE RF requirements for a release independent dual uplink intra-band non-contiguous CA configuration

Clause	Description
5.6A.1	Channel bandwidths per operating band for CA
6.2.2A	UE maximum output power for CA
6.2.3A	UE Maximum Output power for modulation / channel bandwidth for CA
6.2.5A	Configured transmitted Power for CA
6.3.2A	UE Minimum output power for CA
6.3.3A	UE Transmit OFF power for CA
6.3.4A	ON/OFF time mask for CA
6.3.5A	Power control for CA
6.5.1A	Frequency error for CA
6.5.2A	Transmit modulation quality for CA
6.6.1A	Occupied bandwidth for CA
6.6.2.1A	Spectrum emission mask for CA
6.6.2.3	Adjacent Channel Leakage Ratio
6.6.3.1A	Spurious Emission for CA
6.6.3.2A	Spurious emission band UE co-existence for CA
7.3.1A	Reference sensitivity for CA
7.6.2.1A	Out-of-band blocking for CA
7.7.1A	Spurious response for CA

B.4.8 Common UE RF requirements for three uplink inter-band CA configuration

The requirements and test cases listed in Table B.4.8-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.8-1: Common UE RF requirements for a release independent three uplink inter-band CA configuration

Clause	Description
5.6A.1	Channel bandwidths per operating band for CA
6.2.2A	UE maximum output power for CA
6.2.5A	Configured transmitted Power for CA
6.3.2A	UE Minimum output power for CA
6.3.3A	UE Transmit OFF power for CA
6.3.4A	ON/OFF time mask for CA
6.3.5A	Power control for CA
6.5.1A	Frequency error for CA
6.5.2A	Transmit modulation quality for CA
6.6.1A	Occupied bandwidth for CA
6.6.2.1A	Spectrum emission mask for CA
6.6.2.3	Adjacent Channel Leakage Ratio
6.6.3.1A	Spurious Emission for CA
6.6.3.2A	Spurious emission band UE co-existence for CA
6.7.1A	Transmit intermodulation for CA
7.3.1A	Reference sensitivity for CA
7.6.2.1A	Out-of-band blocking for CA
7.7.1A	Spurious response for CA

B.4.9 Common UE RF requirements for operating bands for UE category NB1 and NB2

The requirements and test cases listed in Table B.4.9-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.9-1: Common UE RF requirements for release independent operating bands for UE category NB1

Clause	Description
5.5F	Operating bands for category NB1 and NB2
5.6F	Channel bandwidth for category NB1 and NB2
5.7.1F	Channel spacing for category NB1 and NB2
5.7.2F	Channel raster for category NB1 and NB2
5.7.3F	Carrier frequency and EARFCN for category NB1 and NB2
5.7.4F	TX–RX frequency separation for category NB1 and NB2
6.2.2F	UE maximum output power for category NB1 and NB2
6.2.3F	UE maximum output power for category NB1 and NB2
6.2.5F	Configured transmitted Power for category NB1 and NB2
6.3.2F	UE Minimum output power for category NB1 and NB2
6.3.3F	Transmit OFF power for category NB1 and NB2
6.3.4F	ON/OFF time mask for category NB1 and NB2
6.3.5F	Power Control for category NB1 and NB2
6.5.1F	Frequency error for UE category NB1 and NB2
6.5.2F	Transmit modulation quality for Category NB1 and NB2
6.6.1F	Occupied bandwidth for category NB1 and NB2
6.6.2F	Out of band emission for category NB1 and NB2
6.6.3F	Spurious emission for category NB1 and NB2
6.7.1F	Transmission intermodulation for category NB1 and NB2
7.3.1F	Reference sensitivity for UE category NB1 and NB2
7.4.1F	Maximum input level for category NB1 and NB2
7.5.1F	Adjacent channel selectivity for category NB1 and NB2
7.6.1.1F	In-band blocking for category NB1 and NB2
7.6.2.1F	Out-of-band blocking for category NB1 and NB2
7.7.1F	Spurious response for category NB1 and NB2
7.8.1F	Intermodulation characteristics for category NB1 and NB2

B.4.10 Common UE RF requirements for operating bands for UE category 0

The requirements and test cases listed in Table B.4.10-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.10-1: Common UE RF requirements for release independent operating bands for UE category 0

Clause	Description
5.5E	Operating bands for UE category 0
7.3.1E	Minimum requirements (QPSK) for UE category 0

B.4.11 Common UE RF requirements for operating bands for UE category M1 and M2

The requirements and test cases listed in Table B.4.11-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.11-1: Common UE RF requirements for release independent operating bands for UE category M1 and M2

Clause	Description
5.5E	Operating bands for UE category 0, UE category M1 and M2 and UE category 1bis
5.7.4E	TX–RX frequency separation for category M1 and M2
6.2.2E	UE maximum output power for Category M1 and M2 UE
6.2.3E	UE maximum output power for modulation / channel bandwidth for category M1 and M2
6.2.4E	UE maximum output power with additional requirements for category M1 and M2 UE
6.3.5E	Power control for category M1 and M2
6.5.1E	Frequency error for UE category M1 and M2
6.5.2E	Transmit modulation quality for category M1 and M2
6.6.3.2	Spurious emission band UE co-existence
7.3.1E	Minimum requirements (QPSK) for UE category 0, M1, M2 and 1bis
7.5	Adjacent Channel Selectivity (ACS)
7.6.1	In-band blocking
7.6.2	Out-of-band blocking
7.6.3	Narrow band blocking
7.8.1	Wide band intermodulation

B.4.12 Common UE RF requirements for operating bands for LTE-based V2X operation

The requirements and test cases listed in Table B.4.12-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.12-1: Common UE RF requirements for release independent operating bands for V2X operation

Clause	Description
5.5G	Operating bands for V2X Communication
5.6G	Channel bandwidth for V2X Communication
6.2.2G	UE maximum output power for V2X Commincation
6.2.3G	UE maximum output power for modulation / channel bandwidth for V2X Communication
6.2.4G	UE maximum output power with additional requirements for V2X Communication
6.2.5G	Configured transmitted power for V2X Communication
6.3.2G	UE Minimum output power for V2X Communication
6.3.3G	Transmit OFF power for V2X Communication
6.3.4G	ON/OFF time mask for V2X Communication
6.3.5G	Power Control for V2X Communication
6.5.1G	Frequency error for V2X Communication
6.5.2G	Transmit modulation quality for V2X Communication
6.6.3G	Spurious emission for V2X Communication
7.3.1G	REFSENS requirements (QPSK) for V2X communication
7.4.1G	Maximum input level for V2X communication
7.5.1G	Adjacent Channel Selectivity (ACS) for V2X communication
7.6.1.1G	In-band blocking for V2X communication
7.6.2.1G	Out-of-band blocking for V2X communication
7.7.1G	Spurious response for V2X communication
7.8.1G	Intermodulation characteristics for V2X communication
7.10.1G	Receiver image for V2X communication. (It is only appliacable for intra-band multi-carrier V2X operation)

B.4.13 Common UE RF requirements for four uplink inter-band CA configuration

The requirements and test cases listed in Table B.4.13-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.13-1: Common UE RF requirements for a release independent four uplink inter-band CA configuration

Clause	Description
5.6A.1	Channel bandwidths per operating band for CA
6.2.2A	UE maximum output power for CA
6.2.5A	Configured transmitted Power for CA
6.3.2A	UE Minimum output power for CA
6.3.3A	UE Transmit OFF power for CA
6.3.4A	ON/OFF time mask for CA
6.3.5A	Power control for CA
6.5.1A	Frequency error for CA
6.5.2A	Transmit modulation quality for CA
6.6.1A	Occupied bandwidth for CA
6.6.2.1A	Spectrum emission mask for CA
6.6.2.3	Adjacent Channel Leakage Ratio
6.6.3.1A	Spurious Emission for CA
6.6.3.2A	Spurious emission band UE co-existence for CA
6.7.1A	Transmit intermodulation for CA
7.3.1A	Reference sensitivity for CA
7.6.2.1A	Out-of-band blocking for CA
7.7.1A	Spurious response for CA

B.4.14 Common UE RF requirements for operating bands for UE category 1bis

The requirements and test cases listed in Table B.4.14-1 are specified in TS 36.101 Rel-16 [2].

Table B.4.14-1: Common UE RF requirements for release independent operating bands for UE category 1bis

Clause	Description
5.5E	Operating bands for UE category 1bis
7.3.1E	Minimum requirements (QPSK) for UE category 1bis

Annex C (normative): Common Requirements for 4Rx

C.1 Common UE RF requirements

The requirements and test cases listed in Table C.1-1 are specified in TS 36.101 Rel-16 [2].

Table C.1-1: RF requirements for 4Rx for single band

Clause	Description
7.3	Reference sensitivity power level
7.4	Maximum input level
7.5	Adjacent channel selectivity
7.6	Blocking characteristics
7.7	Spurious response
7.8	Intermodulation characteristics
7.9	Spurious emissions

The requirements and test cases listed in Table C.1-2 are specified in TS 36.101 Rel-16 [2].

Table C.1-2: RF requirements for 4Rx for CA

Clause	Description
6.2.5A	Configured maximum output power
7.3.1A	Reference sensitivity for CA
7.4.1A	Maximum input level for CA
7.5.1A	Adjacent Channel Selectivity for CA
7.6.1.1A	In-band blocking for CA
7.6.2.1A	Out-of-band blocking for CA
7.6.3.1A	Narrow band blocking for CA
7.7.1A	Spurious response for CA
7.8.1A	Wideband intermodulation for CA

C.2 Common UE demodulation and CSI requirements

The requirements and test cases listed in Table C.2-1 are specified in TS 36.101 Rel-16 [2].

Table C.2-1: UE Demodulation and CSI requirements for 4Rx for single band

Clause	Description
8.10.1 (NOTE)	PDSCH
8.10.2	PDCCH/PCFICH
8.10.3	PHICH
8.10.4	ePDCCH
9.9	CSI reporting for 4Rx UE

The requirements and test cases listed in Table C.2-2 are specified in TS 36.101 Rel-16 [2].

Table C.2-2: UE Demodulation and CSI requirements for 4Rx CA/DC

Clause	Description
8.13	Demodulation of PDSCH CA
8.7.9	SDR of FDD CA (4 layer MIMO)
8.7.10	SDR of TDD CA (4 layer MIMO)
8.7.11	SDR of TDD-FDD CA (4 layer MIMO)
8.7.13	SDR of FDD DC (4 layer MIMO)
8.7.14	SDR of TDD DC (4 layer MIMO)
8.7.15	SDR of TDD-FDD DC (4 layer MIMO)
9.1.1.4.2	CSI CA tests for 4Rx UE

Annex D (normative): Common Requirements for performance enhancements for high speed scenario

D.1 Common RRM requirements for performance enhancements for high speed scenario

The requirements and test cases listed in Table D.1-1 are specified in TS 36.133 Rel-16 [3].

Table D.1-1: RRM requirements for performance enhancements for high speed scenario

Clause	Description
4.2	Cell Re-selection
8.1.2.2	E-UTRAN intra frequency measurements in RRC connected state

D.2 Common UE demodulation requirements for performance enhancements for high speed scenario

The requirements and test cases listed in Table D.2-1 are specified in TS 36.101 Rel-16 [2].

Table D.2-1: UE Demodulation requirements for performance enhancements for high speed scenario

Clause	Description
8.2.1.9	FDD PDSCH
8.2.2.9	TDD PDSCH

Annex E (normative): Common Requirements for 8Rx

E.1 Common UE RF requirements

The requirements and test cases listed in Table E.1-1 are specified in TS 36.101 [2].

Table E.1-1: RF requirements for 8Rx

Clause	Description
7.3	Reference sensitivity power level

The requirements and test cases listed in Table E.1-2 are specified in TS 36.101 [2].

Table E.1-2: RF requirements for 8Rx for CA

Clause	Description
7.3.1A	Reference sensitivity for CA

E.2 Common UE demodulation and CSI requirements

The requirements and test cases listed in Table E.2-1 and Table E.2-2 are specified in TS 36.101 [2].

Table E.2-1: UE Demodulation and CSI requirements for 8Rx for single band

Clause	Description
8.14.1	PDSCH
9.12	CSI reporting for 8Rx UE

Table E.2-2: UE Demodulation and CSI requirements for 8Rx CA/DC

Clause	Description
8.14.2	Demodulation of PDSCH CA
8.7.17	SDR of TDD CA (8 layer MIMO)

Annex F (informative): Change history

Table F.1: Change History

Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
11-2009	RP#46	RP-091141				TS36.307 V0.1.0 approved by RAN (Originally in R4-095022)	0.1.0
02-2010	R4#54	R4-100419				For release 9 version, replace sections 4 to 6 as 'Void' and add a new void section as section 7.	0.2.0
03-2010	RP#47	RP-100162				TS36.307 v1.0.0 for approval	1.0.0
03-2010	RP#47	RP-100162				Approved by RAN	9.0.0
09-2010	RP-49	RP-100927	2			CR LTE_TDD_2600_US spectrum band definition additions to TS 36.307 V900	9.1.0
						Correction of section numbering	9.1.1
12-2010	RP-50	RP-101356	008			Band 42 and 43 parameters for UMTS/LTE 3500 (TDD) for TS 36.307	9.2.0
12-2010	RP-50	RP-101361	005			Introduction of L-band in TS 36.307	9.2.0
12-2010	RP-50	RP-101344	016			CR creating the rel-10 of the 36.307 specification	9.3.0
12-2010	RP-50	RP-101356	012			Band 42 and 43 parameters for UMTS/LTE 3500 (TDD) for TS 36.307	9.3.0
12-2010	RP-50					Raised to Rel-10 with no technical change	10.0.0
01-2011						Correction to history table	10.0.1
06-2011	RP-52	RP-110804	015			Add Expanded 1900 MHz Band (Band 25) in 36.307	10.1.0
06-2011	RP-52	RP-110812	022			Add 2GHz S-Band (Band 23) in 36.307 (Rel 10)	10.1.0
09-2011	RP-53	RP-111255	025			Add Band 22 for LTE/UMTS 3500 (FDD) to TS 36.307	10.2.0
03-2012	RP-55	RP-120305	029			Introduction of Band 26/XXVI to TS 36.307	11.0.0
2012-06	RP-56	RP-120789	043			Introduction of CA_1A-19A to TS 36.307	11.1.0
2012-06	RP-56	RP-120793	049			Introduction of APAC700(FDD) into TS 36.307 Rel-11	11.1.0
2012-06	RP-56	RP-120793	053			Introduction of APAC700(TDD) into TS 36.307 Rel-11	11.1.0
2012-06	RP-56	RP-120791	057			Introduction of e850_LB (Band 27) to TS 36.307	11.1.0
2012-09	RP-57	RP-121335	059			Introduction of CA_1A-21A to TS 36.307	11.2.0
2012-09	RP-57	RP-121295	070r1			Relation between EARFCN for overlapping bands with multiple FBI indication	11.2.0
2012-09	RP-57	RP-121338	072			36.307 CR for LTE_CA_B7	11.2.0
2012-09	RP-57	RP-121337	073			TS 36.307 CR for CA_38	11.2.0
2012-09	RP-57	RP-121327	074			Introduction of CA_B7_B20 in 36.307	11.2.0
2012-09	RP-57	RP-121329	075			Introduction of CA band combination Band3 + Band5 to TS 36.307	11.2.0
2012-09	RP-57	RP-121331	076			Introduction of CA_3A-20A to TS 36.307	11.2.0
2012-09	RP-57	RP-121334	077			Add requirements for inter-band CA of B_1-18 in TS36.307	11.2.0
2012-09	RP-57	RP-121333	078			Introduction of CA_8_20 RF requirements into TS36.307	11.2.0
2012-09	RP-57	RP-121324	079			Introduction of CA_B3_B7 in 36.307	11.2.0
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2012-12	RP-58	RP-121882	097			[Rel-11] Introduction of inter-band CA_11-18 into TS36.307	11.3.0
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History

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